Solution Requirements

Date	29 June 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

• Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do).

• Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Functional Requirements:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Applicant Data Input	• Enter personal details (name, age, gender, income type) • Enter financial details (annual income, employment duration) • Enter credit history details (loan amount, overdue days, credit inquiries) • Form validation for mandatory fields and data type checks
FR-2	Credit Card Approval Prediction	• Submit applicant profile for ML model inference • System returns Approved or Rejected prediction instantly • Display prediction confidence / probability score • Show which features most influenced the decision
FR-3	High-Risk Applicant Screening	• Compliance officer uploads batch applicant records (CSV) • Feature engineering pipeline converts payment codes to binary labels • System flags all high-risk applicants as ineligible • Export flagged records report for compliance review
FR-4	Customer Self-Service Eligibility Check	• Prospective customer enters income, employment status, credit history • System predicts approval likelihood before formal bank submission • **D**isplay eligibility result with key contributing factors • Provide guidance on how to improve eligibility score
FR-5	ML Model Training & Management	• Train four classifiers: Logistic Regression, Random Forest, XGBoost, Decision Tree • Compare model performance metrics (accuracy, F1-score, ROC-AUC) • Save the best-performing model as a .pkl file using joblib • Reload saved model at Flask application startup for inference

FR-6	IBM Watson ML Cloud Deployment	• Deploy trained model to IBM Watson Machine Learning service • Expose model as a REST API endpoint for real-time predictions • Enable cloud-based batch prediction for large applicant datasets • Monitor model performance and prediction logs on IBM Cloud

Non-functional Requirements:

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The Flask web interface must be intuitive and easy to use for both bank analysts and non-technical customers. All form fields must include clear labels, placeholder hints, and error messages. Prediction results must be displayed in plain language (Approved / Rejected) with visual indicators
NFR-2	Security	All data submitted through the web form must be validated server-side to prevent injection attacks. Communication between the Flask app and IBM Watson ML must use HTTPS/TLS encryption. IBM IAM (Identity and Access Management) controls must secure Watson ML API credentials. API keys and secrets must be stored in environment variables, never hardcoded.
NFR-3	Reliability	The system must produce consistent and reproducible predictions for identical input profiles. The trained model (.pkl) must be loaded once at app startup and reused for all requests to avoid inconsistency. The IBM Watson ML deployment must include error handling to return a meaningful response on failure.
NFR-4	Performance	The ML model must return a prediction within 200 milliseconds per request under normal load. The pre-trained model is cached in memory at Flask startup, eliminating retraining latency per request. IBM Watson ML REST API supports concurrent real-time and batch predictions for high-throughput scenarios.
NFR-5	Availability	The application deployed on IBM Cloud must target 99.9% uptime using IBM Cloud's managed infrastructure. IBM Watson Machine Learning endpoints are available independently, ensuring prediction service continuity even during Flask app restarts or maintenance windows.
NFR-6	Scalability	The 3-tier architecture (UI -> Flask Logic -> ML/Cloud) allows each layer to be scaled independently. IBM Cloud Kubernetes enables horizontal scaling of the Flask app under increased application load. The IBM Watson ML deployment supports batch prediction for large-scale applicant screening without code changes.